LEVEL UP ACADEMY

Class 10 | CBSE / GSEB | Science

ANSWER KEY & MARKING SCHEME

Most-Expected Questions: Physics (Ch 9–12)

Chapter / Topic: Light · Human Eye · Electricity · Magnetic Effects of Electric Current

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1-MARK QUESTIONS

Q1. What is the SI unit of the power of a lens?
Answer: Dioptre (D).

Q2. What is the focal length of a plane mirror?
Answer: Infinity.

Q3. Which spherical mirror always forms a virtual, erect and diminished image?
Answer: A convex mirror.

Q4. What is the far point of a normal human eye?
Answer: Infinity.

Q5. Which phenomenon makes the clear sky look blue?
Answer: Scattering of light.

Q6. Write the SI unit of resistivity.
Answer: Ohm metre (Ω m).

Q7. Which has more resistance: a 100 W or a 60 W bulb (both rated 220 V)?
Answer: The 60 W bulb. $R = V^2/P$, so 806.7 Ω versus 484 Ω.

Q8. What is the colour of the insulation on the earth wire?
Answer: Green.

2-MARK QUESTIONS

Q9. The magnification produced by a plane mirror is +1. What does this mean?
Answer: The image is the same size as the object ($|m| = 1$), and the + sign shows it is virtual and erect.

Q10. Find the focal length of a lens of power –2.0 D. What type of lens is it?
Answer: $f = 1/P = 1/(–2.0) = –0.5$ m = –50 cm. It is a concave (diverging) lens.

Q11. You want an erect image of an object using a concave mirror of focal length 15 cm. Where should the object be? Will the image be larger or smaller?
Answer: Between the pole and the focus, that is, less than 15 cm from the mirror. The image will be larger than the object (and virtual).

Q12. Why are copper and aluminium wires usually used for transmitting electricity?
Answer: They have very low resistivity, so little energy is wasted as heat. They are also good conductors and fairly cheap (aluminium is light as well).

Q13. How does the resistance of a wire change when (a) its length is doubled and (b) its area of cross-section is doubled?
Answer: (a) It doubles ($R \propto L$). (b) It halves ($R \propto 1/A$).

Q14. Why does a ray of light bend when it passes from air into glass?
Answer: The speed of light is lower in glass (an optically denser medium) than in air. This change in speed at the boundary makes the ray bend towards the normal.

Q15. Why is the tungsten filament of a bulb able to glow without melting? Why are bulbs filled with nitrogen or argon?

Answer: Tungsten has a very high melting point (3380 °C). The chemically inactive gases stop the hot filament from oxidising, so it lasts longer.

Q16. Why is the magnetic field inside a long solenoid said to be uniform?

Answer: Inside the solenoid the field lines are parallel straight lines, which shows the field has the same strength and direction at all points.

3-MARK QUESTIONS

Q17. An object 4 cm high is placed 25 cm in front of a concave mirror of focal length 15 cm. Find the position, size and nature of the image.

Answer: u = –25 cm, f = –15 cm. 1/v = 1/f – 1/u = –1/15 + 1/25 = –2/75, so v = –37.5 cm (in front of the mirror). m = –v/u = –(–37.5)/(–25) = –1.5, so h′ = –6 cm. The image is real, inverted and magnified (6 cm tall).

Q18. An object is placed 12 cm in front of a convex mirror of radius of curvature 30 cm. Find the position and nature of the image.

Answer: f = R/2 = +15 cm, u = –12 cm. 1/v = 1/f – 1/u = 1/15 + 1/12 = 9/60, so v ≈ +6.7 cm (behind the mirror). m = –v/u ≈ +0.56. The image is virtual, erect and diminished.

Q19. A 6 Ω and a 3 Ω resistor are joined in parallel, and this combination is connected in series with a 2 Ω and a 3 Ω resistor across a 12 V battery….

Answer: Parallel part: 1/R = 1/6 + 1/3 = 1/2, so R = 2 Ω. Total = 2 + 2 + 3 = 7 Ω. I = 12/7 ≈ 1.7 A.

Q20. An electric motor takes 5 A from a 220 V line. Find its power and the energy it uses in 2 hours.

Answer: P = VI = 220 × 5 = 1100 W. E = 1.1 kW × 2 h = 2.2 kWh (= 7.92×10^6 J).

Q21. The hot plate of an electric oven on a 220 V line has two coils A and B of 24 Ω each, which can be used separately, in series or in parallel. Find…

Answer: Separately: I = 220/24 ≈ 9.2 A. In series: R = 48 Ω, I = 220/48 ≈ 4.6 A. In parallel: R = 12 Ω, I = 220/12 ≈ 18.3 A.

Q22. A copper wire 1 km long has a cross-sectional area of 0.5 mm^2. Find its resistance. (ρ of copper = 1.62×10^{-8} Ω m)

Answer: A = 0.5 mm^2 = 5×10^{-7} m^2. R = ρL/A = $(1.62 \times 10^{-8} \times 1000)/(5 \times 10^{-7})$ ≈ 32.4 Ω.

5-MARK QUESTIONS

Q23. (a) How would you connect three resistors of 6 Ω each to get a total resistance of (i) 9 Ω and (ii) 4 Ω? [3] (b) Several 10 W bulbs are designed for…

Answer: (a) (i) Two in parallel (3 Ω), then the third in series: 3 + 6 = 9 Ω. (ii) Two in series (12 Ω), then that pair in parallel with the third: 1/R = 1/12 + 1/6 = 1/4, so R = 4 Ω. (b) Maximum power = VI = 220 × 5 = 1100 W. Number of bulbs = 1100/10 = 110.

Q24. (a) Draw ray diagrams for a convex lens with the object (i) at $2F_1$ and (ii) between F_1 and O. State the nature and position of the image in each…

Answer: (a) (i) The image is at $2F_2$: real, inverted, same size. (ii) The image is on the same side as the object: virtual, erect, enlarged. Use two rays in each: one parallel to the axis that passes through F_2, and one through O undeviated. (b) When the object is placed between F_1 and the optical centre, a convex lens forms a virtual, erect and enlarged image, so small things look bigger.

Q25. (a) What is myopia? Explain its causes and correction. [3] (b) Why does the Sun appear reddish at sunrise and sunset? [2]

Answer: (a) A person with myopia can see near objects clearly but not distant ones, because the image forms in front of the retina. Causes: excessive curvature of the eye lens, or elongation of the eyeball. Correction: a concave lens of suitable power. (b) Near the horizon, sunlight travels a longer

path through the atmosphere. Most of the blue light is scattered away, so mainly red light, which is scattered least, reaches our eyes.